

Creating Microservices for account and loan

```
> main > java > com > cognizant > account > controller > AccountController.java > Language Support for Java(TM) by Red Hat > {} com.cognizant.a

1  package com.cognizant.account.controller;
2
3  import org.springframework.web.bind.annotation.GetMapping;
4  import org.springframework.web.bind.annotation.PathVariable;
5  import org.springframework.web.bind.annotation.RequestMapping;
6  import org.springframework.web.bind.annotation.RestController;
7
8  import java.util.LinkedHashMap;
9  import java.util.Map;
10
11  @RestController
12  @RequestMapping("/accounts")
13  public class AccountController {
14
15      @GetMapping("/{number}")
16      public Map<String, Object> getAccount(@PathVariable String number) {
17          Map<String, Object> account = new LinkedHashMap<>();
18          account.put(key: "number", number);
19          account.put(key: "type", value: "savings");
20          account.put(key: "balance", value: 234343);
21          return account;
22      }
23  }

account > src > main > java > com > cognizant > account > AccountApplication.java > Language Support for Java(TM) by Red Hat >

1  package com.cognizant.account;
2
3  import org.springframework.boot.SpringApplication;
4  import org.springframework.boot.autoconfigure.SpringBootApplication;
5
6  @SpringBootApplication
7  public class AccountApplication {
8      Run | Debug
9      public static void main(String[] args) {
10          SpringApplication.run(AccountApplication.class, args);
11      }
12  }
```

```

account > src > test > java > com > cognizant > account > J AccountControllerTest.java > {} com.cognizant.account
1  package com.cognizant.account;
2
3  import org.junit.jupiter.api.Test;
4  import org.springframework.beans.factory.annotation.Autowired;
5  import org.springframework.boot.test.autoconfigure.web.servlet.AutoConfigureMockMvc;
6  import org.springframework.boot.test.context.SpringBootTest;
7  import org.springframework.test.web.servlet.MockMvc;
8
9  import static org.springframework.test.web.servlet.request.MockMvcRequestBuilders.get;
10 import static org.springframework.test.web.servlet.result.MockMvcResultMatchers.jsonPath;
11 import static org.springframework.test.web.servlet.result.MockMvcResultMatchers.status;
12
13 @SpringBootTest
14 @AutoConfigureMockMvc
15 class AccountControllerTest {
16
17     @Autowired
18     private MockMvc mockMvc;
19
20     @Test
21     void shouldReturnAccountDetails() throws Exception {
22         mockMvc.perform(get("/accounts/00987987973432"))
23             .andExpect(status().isOk())
24             .andExpect(jsonPath("$.number").value("00987987973432"))
25             .andExpect(jsonPath("$.type").value("savings"))
26             .andExpect(jsonPath("$.balance").value(234343));
27     }
28 }

```

```

loan > src > main > java > com > cognizant > loan > controller > J LoanController.java > Language Support for Java(TM) by Red Hat
1  package com.cognizant.loan.controller;
2
3  import org.springframework.web.bind.annotation.GetMapping;
4  import org.springframework.web.bind.annotation.PathVariable;
5  import org.springframework.web.bind.annotation.RequestMapping;
6  import org.springframework.web.bind.annotation.RestController;
7
8  import java.util.LinkedHashMap;
9  import java.util.Map;
10
11 @RestController
12 @RequestMapping("/loans")
13 public class LoanController {
14
15     @GetMapping("/{number}")
16     public Map<String, Object> getLoan(@PathVariable String number) {
17         Map<String, Object> loan = new LinkedHashMap<>();
18         loan.put(key: "number", number);
19         loan.put(key: "type", value: "car");
20         loan.put(key: "loan", value: 400000);
21         loan.put(key: "emi", value: 3258);
22         loan.put(key: "tenure", value: 18);
23         return loan;
24     }
25 }
26

```

```

loan > src > main > java > com > cognizant > loan > J LoanApplication.java > Language Support for Java(TM) by Red Hat > {} com.cognizant.loan
1  package com.cognizant.loan;
2
3  import org.springframework.boot.SpringApplication;
4  import org.springframework.boot.autoconfigure.SpringBootApplication;
5
6  @SpringBootApplication
7  public class LoanApplication {
8      Run | Debug
9      public static void main(String[] args) {
10         SpringApplication.run(LoanApplication.class, args);
11     }
12 }

```

```

loan > src > test > java > com > cognizant > loan > J LoanControllerTest.java > {} com.cognizant.loan
1  package com.cognizant.loan;
2
3  import org.junit.jupiter.api.Test;
4  import org.springframework.beans.factory.annotation.Autowired;
5  import org.springframework.boot.test.autoconfigure.web.servlet.AutoConfigureMockMvc;
6  import org.springframework.boot.test.context.SpringBootTest;
7  import org.springframework.test.web.servlet.MockMvc;
8
9  import static org.springframework.test.web.servlet.request.MockMvcRequestBuilders.get;
10 import static org.springframework.test.web.servlet.result.MockMvcResultMatchers.jsonPath;
11 import static org.springframework.test.web.servlet.result.MockMvcResultMatchers.status;
12
13 @SpringBootTest
14 @AutoConfigureMockMvc
15 class LoanControllerTest {
16
17     @Autowired
18     private MockMvc mockMvc;
19
20     @Test
21     void shouldReturnLoanDetails() throws Exception {
22         mockMvc.perform(get("/loans/H00987987972342"))
23             .andExpect(status().isOk())
24             .andExpect(jsonPath("$.number").value("H00987987972342"))
25             .andExpect(jsonPath("$.type").value("car"))
26             .andExpect(jsonPath("$.loan").value(400000))
27             .andExpect(jsonPath("$.emi").value(3258))
28             .andExpect(jsonPath("$.tenure").value(18));
29     }
30 }

```

```

[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 5.457 s -- in com.cognizant.loan.LoanControllerTest
[INFO] Results:
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 13.073 s
[INFO] Finished at: 2026-07-20T16:11:51+05:30
[INFO] -----

```

```
1  {  
2    "number": "00987987973432",  
3    "type": "savings",  
4    "balance": 234343  
5  }
```

```
1  {  
2    "number": "H00987987972342",  
3    "type": "car",  
4    "loan": 400000,  
5    "emi": 3258,  
6    "tenure": 18  
7  }
```